

# Components

Currently in possession:

CAD:

- PLA+ Filament
- need - TPU if the roller doesn't work

Electronics:

| Component name                                                                          | Description                                                                                                                                                                                                                                                                                  | Nr | Order link (amazon or ali express)                                                                                                                                                                                | Wired - yes/no                                                               | Need to order more? - yes/no, how many? |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------|
| Motor - N20                                                                             | Type - right-angled, comes with support for pinning<br>- Specifications:<br>- Voltage: DC 12 V;<br>- No-Load Speed: 27 RPM;<br>- Gearbox Size ( L × W × H ): 0.71 × 0.47 × 0.39 inch / 18 × 12 × 10 mm;<br>- Shaft Size ( D × H ): 0.12 × 0.3 inch / 3 × 7.5 mm.<br>Additional: Supports - 2 | 1  | <a href="https://www.amazon.co.uk/dp/B0DPKZ3C7C?ref=ppx_yo2ov_dt_b_fed_asin_title&amp;th=1">https://www.amazon.co.uk/dp/B0DPKZ3C7C?ref=ppx_yo2ov_dt_b_fed_asin_title&amp;th=1</a>                                 | Yes - but needs more soldering or wire cover                                 | TBD                                     |
| Motor - N20 with 100cm threaded shaft<br>Nut for it should arrive until the 31st of May | Type - right-angled, with long threaded shaft<br>Specifications:<br>- Speed: 500RPM<br>- Voltage: 12V, 6V;<br>- Horsepower: 2Watts;<br>- Weight: 19 grams;<br>- Material: metal;<br>- Full length: 125mm                                                                                     | 2  | <a href="https://www.amazon.co.uk/gp/product/B07HQPVPK/ref=ox_sc_act_title_6?smid=A2LZADUCQG7RBO&amp;psc=1">https://www.amazon.co.uk/gp/product/B07HQPVPK/ref=ox_sc_act_title_6?smid=A2LZADUCQG7RBO&amp;psc=1</a> | No - needs wire soldered. Requires nut that spins around the shaft - ordered | No - ordered some with longer shaft     |
| Motor - DC TT                                                                           | Yellow Motor for Mecanum Wheels                                                                                                                                                                                                                                                              | 4  | <a href="https://www.amazon.co.uk/dp/B09BYWXBNC?ref=ppx_yo2ov_dt_b_fed_asin_title&amp;th=1">https://www.amazon.co.uk/dp/B09BYWXBNC?ref=ppx_yo2ov_dt_b_fed_asin_title&amp;th=1</a>                                 | Yes. Requires support - to order                                             | Yes - need one more pack for backup     |
| Should                                                                                  | Type: Right-angled motor                                                                                                                                                                                                                                                                     | 2  | <a href="https://www.amazon">https://www.amazon</a>                                                                                                                                                               | No                                                                           | TBD (No)                                |

|                                                                                 |                                                         |  |                                                                                                                                                                  |  |  |
|---------------------------------------------------------------------------------|---------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <b>arrive on the first week of June:</b><br>Motor - N20 with long shaft - 160mm | with threaded shaft of 160 mm. Motor dimensions = 25mm. |  | <a href="https://www.amazon.co.uk/dp/B0GV5G5SGV?ref=ppx_yo2ov_dt_b_fed_asin_title&amp;th=1">n.co.uk/dp/B0GV5G5SGV?ref=ppx_yo2ov_dt_b_fed_asin_title&amp;th=1</a> |  |  |
|---------------------------------------------------------------------------------|---------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

| Component name | Description     | Nr | Order link (amazon or ali express)                                                                                                                                                                                                      | Soldered with the pins - yes/no | Need to order more? - yes/no, how many? |
|----------------|-----------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------|
| Motor driver   | Type: TB6612FNG | 5  | <a href="https://www.aliexpress.com/item/1005010405309752.html?spm=a2g0o.order_list.order_list_main.58.421f1802ZVMVIF">https://www.aliexpress.com/item/1005010405309752.html?spm=a2g0o.order_list.order_list_main.58.421f1802ZVMVIF</a> | No                              | TBD (No)                                |
| Motor driver   | Type: L298N     | 2  | <a href="https://www.amazon.co.uk/dp/B07XJSM LGQ?ref=ppx_yo2ov_dt_b_fed_asin_title">https://www.amazon.co.uk/dp/B07XJSM LGQ?ref=ppx_yo2ov_dt_b_fed_asin_title</a>                                                                       | Yes                             | TBD (No)                                |

| Component name         | Description                                                                                                                                                                     | Nr | Order link (amazon or ali express)                                                                                                                                                                                                                                                                                                                                                                                   | Soldered with the pins - yes/no | Need to order more? - yes/no, how many? |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------|
| Buck Converter : Dc-Dc | 1. Ali-express:<br>DC-DC Buck Converter Module<br>Buck Regulator<br>6-40V to 1.2-36V<br>Output Voltage Adjustable Converter<br>Power Supply Module 300W 20A CC<br><br>2. Amazon | 1  | 1. <a href="https://www.aliexpress.com/item/1005009863574561.html?spm=a2g0o.order_list.order_list_main.11.421f1802ZVMVIF">https://www.aliexpress.com/item/1005009863574561.html?spm=a2g0o.order_list.order_list_main.11.421f1802ZVMVIF</a><br><br>2. <a href="https://www.amazon.co.uk/dp/B081X5YX8V?ref=ppx_yo2ov_dt_b_fed_asin_title">https://www.amazon.co.uk/dp/B081X5YX8V?ref=ppx_yo2ov_dt_b_fed_asin_title</a> | N/A                             | TBD (No)                                |

ESP32 - 1 piece  
Terminal board for esp32 - 1 piece  
Wire housing/melting cover - 1 box  
Breadboards - 4 small  
AA battery pack  
AA battery holders - 3x4AA  
Lipo batteries - 2S 7.4V 3500A 35CC  
Lipo charger  
Mecanum wheels x4  
Gear for N20 shaft - 1x 4 sizes

Yet to arrive: nut for 100mm motor shaft & 160mm N20 motor

To order: (1) Perfboard & terminals or (2) PCB designed